

E-Commerce Product Entity Resolution

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Abstract

The **E-Commerce Product Entity Resolution & Matching Platform** is a DSA-based project designed to identify and group identical products listed by different online sellers. In e-commerce websites, the same product often appears with different names, descriptions, and prices, making it difficult for customers to compare options. This system solves the problem by normalizing product information and matching products based on attributes such as brand, model, storage, color, and other specifications. It uses efficient data structures like **HashMap**, **HashSet**, **ArrayList**, and searching and sorting algorithms to detect duplicate products and create a single standardized product entry. The platform also provides price comparison across multiple sellers, helping users find the best available deal. The project demonstrates practical applications of Data Structures and Algorithms in solving real-world e-commerce challenges. It is simple to implement using Java without requiring machine learning or a database, making it an ideal mini-project for students while improving their understanding of searching, hashing, sorting, and string matching techniques.

Introduction

- E-commerce platforms often contain multiple listings for the same product.
- Different sellers may use different names, formats and descriptions.
- This makes product comparison and catalog management difficult.
- The proposed system resolves duplicate product entities and creates a standardized product group.
- It also helps users compare seller prices efficiently.

Software Requirements

Development Tools:

VS Code / IntelliJ IDEA

JDK 17 or later

Git & GitHub

DSA Technologies:

HashMap

HashSet

ArrayList

String Matching

Searching and Sorting Algorithms

Hardware Requirements

- Processor: Intel Core i3 or equivalent
- RAM: Minimum 4 GB; 8 GB recommended
- Storage: At least 500 MB free space
- Display: 1366 × 768 or higher
- Keyboard and mouse
- Internet connection: Optional for collecting sample datasets

Literature Survey

- Brizan & Tansel (2006): surveyed entity resolution and record linkage methods for data cleansing and integration.
- Christen (2012): described a general data-matching workflow including preprocessing, indexing, comparison and evaluation.
- Köpcke et al. (2010): compared entity-matching frameworks and emphasized blocking and combining matching algorithms.
- Macková & Pilát (2024/2025): studied product mapping datasets and methods for matching product listings across e-shops.
- Recent research shows that product matching remains challenging because product data is heterogeneous.

Methodology

- 1. Collect product records from multiple sellers.
- 2. Normalize names: lowercase text, remove unnecessary symbols and spaces.
- 3. Extract important attributes such as brand, model, storage and color.
- 4. Compare records using exact and string-similarity rules.
- 5. Use a threshold to classify records as same or different products.
- 6. Store matched groups using HashMap and remove duplicates using HashSet.
- 7. Sort seller listings by price and display the lowest price.

Conclusion

- The project provides a simple DSA-based solution for product matching.
- It reduces duplicate product listings and organizes seller information.
- It demonstrates practical use of hashing, searching, sorting and string matching.
- The system can be extended later with fuzzy matching, databases or machine learning.

THANK YOUU!!